SOLIU ADEMOLA MUDASHIRU

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RESEARCH SUMMARY

Theorizing, understanding and solving earth system dynamics through data driven modelling with a view to making informed decision for sustainable ecosystems.

Research Interests: Groundwater resources management and environmental hydrogeology; GIS, remote sensing, and geospatial modelling; Machine learning and AI approaches in geoscience; Environmental geophysics and subsurface characterization; Natural resources assessment and ecosystem sustainability; Policy and decision making.

EDUCATION

Master of Technology (M.Tech) in Exploration Geophysics (4.43/5.00 or 88.6%)

Federal University of Technology, Akure (FUTA) 2025

Specialization: Groundwater, Environmental, Engineering, Mining

Bachelor of Technology (B.Tech) in Applied Geophysics (3.82/5.00 or 76.4%)

Federal University of Technology, Akure (FUTA) 2018

KEY SKILLS

Programming languages – (Python, R), **GIS – (**ArcGIS Desktop, ArcGIS Online, Google Earth Engine), **Remote Sensing – (**Experience with Sentinel 1 & 2, SRTM, ALOS- PALSAR, LiDAR, and ASTER), **Modelling –** (Hydrologic, Landfill, Ecosystem, Urban green), **Data Analysis – (**Python, R, Excel, PowerBI), **Machine Leaning and AI** (Sci-Kit Learn, PyTorch), **Soft Skills –** (Strong writing, teamwork and independent research abilities).

WORK EXPERIENCE

Teaching Assistant – Department of Applied Geophysics

Federal University of Technology, Akure 2026 - Present

- Assisting in undergraduate instructions and laboratory sessions.
- Supporting field based geophysical training exercises.
- Guiding students in data interpretation and technical report writing.
- Contributing to exam supervision, grading and academic mentoring.

Environmental Scientist – Mukolak GeoConsult

Nigeria Limited, Akure 2023– Present

- Assisting in the coordination of researchers in geophysical field surveys for groundwater and environmental studies.
- Supporting on-site measurements of relevant water parameters like EC, pH, water table depth, and discharge measurement.
- Collaborating in technical report writing.

Research Assistant - Department of Applied Geophysics

Federal University of Technology, Akure 2021 – 2025

- Conducted groundwater flow modelling and aquifer characterization using GIS-based analytical tools.
- Interpreted hydrogeological parameters to evaluate groundwater potential.
- Prepared groundwater monitoring datasets, processed time-series data, and contributed to field data acquisition.
- Developed maps for groundwater vulnerability, recharge zones, and hydraulic properties using ArcGIS & Google Earth Engine.
- Mentored three final year undergraduate students in their thesis data collection and writing.

GIS & Remote Sensing Expert – Freelance

Nigeria 2022 – Present

- Processing multispectral imagery from Sentinel and Landsat tools for land cover mapping and groundwater related studies.
- Carrying out spatial interpolation and geostatistical analysis using ArcGIS and Python programming language.
- Producing thematic maps and digital elevation analyses to support environmental assessments

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RESEARCH EXPERIENCE

"Artificial intelligence (AI) for flood vulnerability assessment" 2026

- Analyzing EO data to determine flood conditioning parameters like runoff generation indicators
- Employing the effectiveness of Google Earth Engine (GEE) to spatially assess areas of flood occurrence in the study area
- Evaluating ML algorithms to predict flood vulnerability of the studied region

"Machine learning frameworks and data driven modelling for Copper prospectivity" 2025

- I conceptualized the adopted conditioning factors.
- I designed the suitable ML algorithms and Entropy model used for the prospectivity prediction.
- I co-authored the submitted Team slide for the Next Generation Explorers Awards (NGEA) submission.

"Urban groundwater modelling leveraging on Fuzzy logic approach" 2025

- I designed fuzzy-based hydrologic model integrating factors from remote sensing and field data acquisition.
- I conceptualized urban water quality indicators (pH, TDS) to predict vegetation vulnerability.
- I drafted the accepted PhD proposal from this research.

"Geospatial technology-based MCDA and machine learning algorithms: an ensemble and inter-evaluating frameworks for conceptualizing groundwater potential" 2025

- I processed and analyzed all the ecological modelling datasets (LULC, DEM, PERSIANN).
- I performed all the machine learning modelling using Python programming language.
- I produced the water availability maps based on the models using ArcGIS 10.7 software.

"Conceptualization of MEREC model for groundwater potential deciphering leveraging on geophysical datasets in a basement terrain" 2025

- I coded the Methods Based on Removal Effects of Criteria using Python programming language for the assessment.
- I conceptualized the in-situ geophysical datasets (Aquifer resisitivity, Reflection coefficient etc.) which were employed as modelling inputs.
- I produced the groundwater partitioning map using ArcGIS 10.8 software.

"Prediction of groundwater vulnerability in a basement complex area employing GIS based comparative MCDMs approach" 2024

- I conceptualized the Multi Criteria Decision Making Models for the groundwater vulnerability assessment.
- I integrated remote sensing data (i.e. DEM, PERSIANN) with geophysical datasets for the groundwater vulnerability assessment.
- I wrote the accepted proposal.

FIELD WORK EXPERIENCE

"Hydrogeophysics survey for borehole development" 2026

- I set up the R-50 resistivity equipment to acquire the hydrogeophysics data in the study area
- I analyzed and interpreted the collected horizontal profiling and Schlumberger data with the other team members.
- I wrote the report detailing the findings.

"Hydrogeologic Modelling Survey" Oba – Ile, Akure, Nigeria 2025

- I set up the PASSI resistivity equipment to acquire the 1D VES data in the study area
- I analyzed and interpreted the collected data with the other team members.
- I wrote the report detailing the findings.

"Borehole Development Geophysical Survey" Awo-Ekiti, Ekiti State, Nigeria 2025

- I collaborated in the 1D VES data acquisition and the instrumental setup for the data gathering
- I analyzed and interpreted the collected data with the other team members.
- I wrote the report detailing the findings.

"Application of MEREC model for Water Analysis" Akure, Nigeria 2025

- I led the graduate student in the field data collection of well depth and static water level data in the study area
- I configured the navigation instrument for smooth well location data acquisition.
- I gave the graduate students insights into the computations of the well depth data

"Iron Ore Deposit Mapping at Ebiya Ajaokuta Local Government", Kogi state, Nigeria 2023

- I conducted Vertical Electrical Sounding (VES) data acquisition with the other team members.
- I interpreted VES results using WinResist software to locate conductive zones diagnostic of iron ore occurrence.
- I assisted in the curation of magnetic data interpreted to delineate the magnetic ore body from the surrounding geologic formations

"Geophysical survey for subsurface characterization" Igarra, Edo State, Nigeria 2023

- I interpreted the 2D ERT data acquired using DIPRO^{TM} inversion software to show lateral and vertical visualization of the geologic settings of the study site
- I interpreted the ground truth magnetic data to locate geologic contact zones as well as structures that could be inimical to the foundation design of the study site

PUBLICATIONS AND REPORTS

Published

- Ozegin, K.O., **Mudashiru, S.A** & Ejepu, J.S (2026). Geospatial technology-based MCDA and machine learning algorithms: an ensemble and inter-evaluating frameworks for conceptualizing groundwater potential. *Ecological Frontiers (Elsevier). Link*
- Mogaji, K.A., **Mudashiru, S.A**., Oguntade, S.S & Toyib, M. Y (2026). Geospatial Modelling of Landfill Suitability Using Geophysical and Remote Sensing Data in a Basement Complex Area. *Discover Applied Sciences (Springer Nature). Link*
- **Mudashiru, S.A**., Olatunji, A.O., Oke, P.A, Adeyanju, O.L., Orewale, T.P (2024). Subsurface competence evaluation using electrical resistivity method at a proposed building site along FUTA staff quarters, Oba Nla, Akure Southwestern Nigeria. *Pakistan Journal of Geology, 8 (1): 80.87. Link*

Under Review

- Mogaji, K.A., Eval, J.F., Afolabi, D.O., **Mudashiru, S.A.,** Effiong, A.J & Adegoke, Q.O (2025). Landslide susceptibility assessment using Catastrophe Theory and Analytical Hierarchy Process Concepts in Southwestern Nigeria: Insights from geophysical, geological and meteorological data sets. *Earth Systems and Environment (Springer Nature).*
- Mogaji, K.A., **Mudashiru, S.A.,** Ozegin, K.O & Oguntade, S.S (2025). Groundwater potential assessment leveraging the hybrid objective model (IDOCRIW-CoCoSo) in the basement terrain of Nigeria: insights from remote sensing and geophysical datasets. *Cleaner Water (Elsevier).*
- **Mudashiru, S.A.,** Mogaji, K.A & Ozegin, K.O (2025). Modelling groundwater vulnerability leveraging a developed Python-coded IDOCRIW-MAUT model in a heterogeneous geologic environment of Nigeria. *Geosystems and Geoenvironment (Elsevier).*
- Mogaji, K.A & **Mudashiru, S.A** (2024). Groundwater potentiality assessment applying python-based IDOCRIW-COCOSO objective modelling algorithm in a multi-faceted geologic environment of Nigeria. *HydroResearch (KeAi & Elsevier). Preprint Online*

Submission proof on my github repository

Co-Authored Technical Reports

Link to reports on my github repository

OTHER COMPETENCIES

"Critical Writing Expert at Teflon Workshop" Remote | Nigeria 2025– Present

- Developed tailored academic reports, essays, and research-based assignments for diverse clients
- Transformed complex topics into insightful documents that showcase high-level of critical reasoning

"Project Work Reviewer", Federal University of Technology, Akure, Nigeria 2023– Present

- I leveraged my research ability to review undergraduate project write up for publications
- I offered insights for graduate research work submission into reputable journals
- I review graduate students thesis prior to submission to their supervisors

"Machine Learning Expert at Infinity Research Consultant" Remote | UK 2022 – 2024

- I developed machine learning models (i.e. XGBoost, Random Forest, SVM etc.) for diverse projects.
- I led data analysis projects across several team members.

THESIS

Master's Thesis

"Development of IDOCRIW-COCOSO and IDOCRIW – MAUT Decision Support System Models for Groundwater Potentiality and Vulnerability in a Basement Terrain" 2025

Bachelor's Thesis

"Subsoil evaluation for pre-foundation study using geophysical methods at FUTA staff quarters" 2018

CONFERENCES AND WEBINARS

Second Sub-Saharan African Regional Virtual Conference by SEG	2024
The Role of Geophysics in HDD Webinar by SEG FUTA	2023
Water and Development Symposium Virtual Conference by IHE DELFT	2023
3rd World Symposium on Climate Change Adaptation (WSCCA- 2017), FUTA	2019

REFERENCES

Available upon request